HOW A GENERATOR MAKES ELECTRICITY

Almost every power plant ends the same way: a spinning magnet inside a coil of copper.

1

SPIN

Steam, hot gas, an engine, water or wind turns a shaft.

2

MAGNET IN A COIL

The shaft spins a magnet (the rotor) inside copper windings (the stator).

3

CURRENT

The moving magnetic field pushes current through the copper: alternating current, 60 times a second.

60 Hz

US grid frequency: every generator on it spins in step

3,600 rpm

a two-pole gas or steam turbine generator at 60 Hz

21 kV

SW-3X1's generators; the step-up transformer lifts it to 230 kV

Copper

the windings, the bus, the cable: all of it carries the current out

Solar panels, fuel cells and batteries are the exceptions: they make DC, and an inverter turns it into 60 Hz AC for the grid.

FROM THE GENERATOR TO THE LIGHT SWITCH

Power is made at medium voltage, raised to high voltage to travel, then stepped down again near the customer.

1

GENERATOR

13.8–25 kV

21 kV in SW-3X1

CABLE

isolated-phase bus, MV-105

›

2

STEP-UP (GSU)

→ 115–765 kV

230 kV in SW-3X1

CABLE

HV XLPE cable, switchyard bus

›

3

TRANSMISSION

115–765 kV

long distance, overhead

CABLE

overhead conductor, OPGW

›

4

SUBSTATION

→ 4–35 kV

steps it down for the town

CABLE

substation cable, grounding

›

5

DISTRIBUTION

4–35 kV

feeders along the streets

CABLE

MV URD, overhead distribution

›

6

CUSTOMER

480 / 240 / 120 V

factory, office, home

CABLE

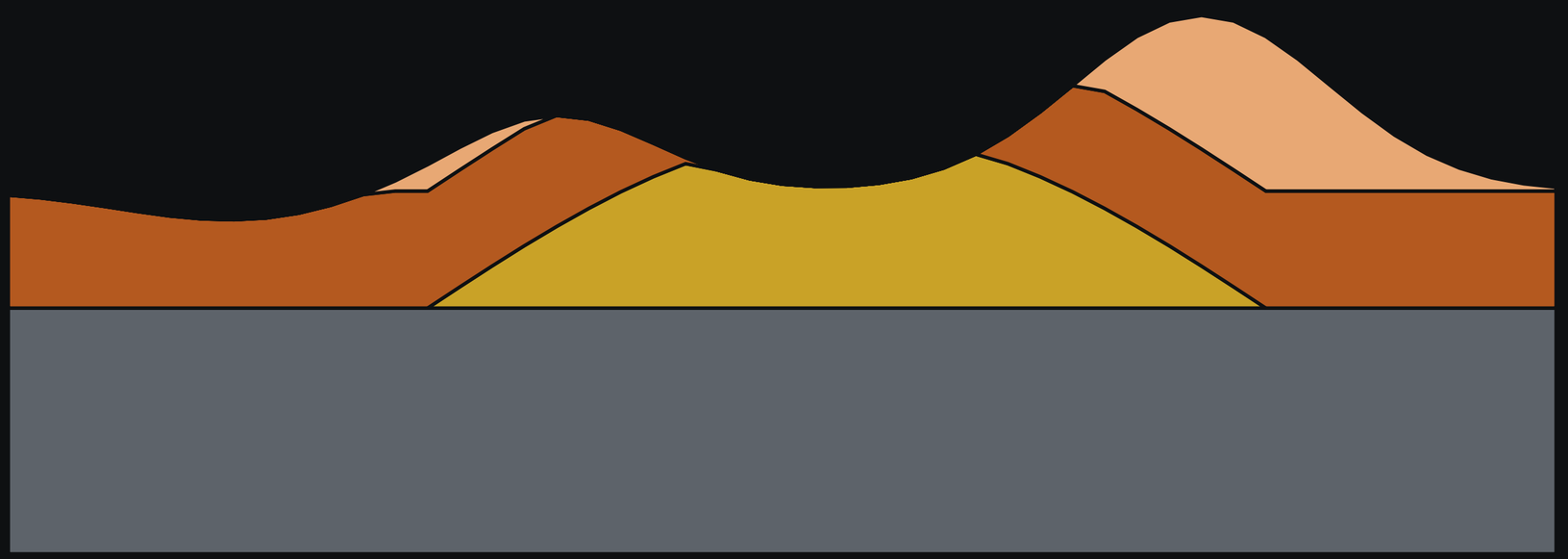
building wire, MC, service entrance

← INSIDE THE PLANT: TODAY

Higher voltage, lower current, smaller losses: that is why transmission runs at hundreds of kV. Southwire has a product family at every step, from the generator to the outlet.

A DAY ON THE GRID: WHO RUNS WHEN

Demand rises in the morning and peaks in the evening. Each kind of plant takes a layer.



- Peaking: simple cycle, engines, batteries
- Mid-merit: combined cycle
- Solar
- Baseload: nuclear, coal, combined cycle

The evening peak arrives as the sun sets: that is when gas and batteries earn their keep.

Illustrative summer weekday, not a real grid's data.

Every start and stop cycles the plant, and its cable: heat, vibration and terminations that move. Peakers may start every day.

FRONT OF THE METER, BEHIND THE METER

The same machines, two business models. The meter is the line between the utility and the customer.

FRONT OF THE METER

THE UTILITY-SCALE PLANT

- Sells power into the grid through a switchyard and the POI
- Utilities and IPPs; the ISO dispatches it
- HV gen-tie, switchyard, GSUs: the big cables
- Most of SW-3X1's power block

BEHIND THE METER

POWER AT THE CUSTOMER'S SITE

- Feeds a load directly: data centre, factory, campus
- Often modular engines, turbines, fuel cells and batteries
- MV collection and LV distribution on site; little or no grid tie
- SW-3X1's data-centre campus and modular yard

Behind-the-meter is the fastest-growing lane: data centres that cannot wait years for a grid connection build their own power.

THE CABLE FAMILIES IN A POWER PLANT

From the biggest to the smallest. The whole SW-3X1 campus: 21,217 cables, 1,212 miles.

HV — 69–345 kV
XLPE: switchyard and gen-tie

MV — 5–35 kV
MV-105: big motors, transformers, feeders

LV POWER — 600 V
TC-ER, MC, ARMOR-X: motors, panels, lighting

VFD — 600 V–2 kV
drive-rated: fans and pumps on drives

CONTROL — 600 V
TC-ER multi-conductor: breakers, valves, interlocks

INSTRUMENTATION — 300 V
PLTC / ITC shielded pairs, thermocouple extension

FIBRE — –
controls network, protection

GROUNDING — bare Cu
the ground grid under everything

By length: Instrumentation 44% · LV 600 V power 20% · Control 18% · Fibre, fire alarm, other 11% · MV / HV 7%. The big cables get the attention; the small ones make up most of the miles.

Shares by footage from the SW-3X1 model cable schedule; voltage ratings typical.

stephan.hardt@southwire.com

WHAT A POWER PLANT DOES TO A CABLE

STRESS	WHERE	THE ANSWER
HEAT	beside turbines, HRSGs and steam lines	high-temperature ratings, route away from hot surfaces
CYCLING	daily starts: heat, vibration, moving terminations	flexible strands, good terminations, support
ELECTRICAL NOISE	variable-frequency drives on fans and pumps	VFD-rated, shielded cable
HAZARDOUS AREAS	gas skids and fuel yards (Class I, Div 2)	ARMOR-X MC-HL, no conduit
FIRE AND SMOKE	halls, control rooms, cable spreading rooms	flame-retardant, low-smoke, circuit integrity
OUTDOORS	sun, oil, water, burial	sunlight- and oil-resistant jackets, direct-burial ratings
SCHEDULE	long pulls on the critical path	SIMpull reels, cut-to-length, pre-terminated

The wrong cable fails early, and a cable failure can trip a unit. That is why the spec is strict, and why it is written for each area of the plant.

stephan.hardt@southwire.com

GAS PLANTS ARE GETTING CLEANER, AND MORE CABLE-DENSE

HYDROGEN

- Gas turbines can burn a blend of hydrogen with natural gas; OEMs are working toward 100%
- Our model has an H2 receiving station, trailer bay and an electrolyzer building
- More gas-detection, hazardous-area and DC cable

CARBON CAPTURE

- Captures CO2 from the stack: absorber, stripper, compressors
- A second plant's worth of motors, drives and instruments beside the first
- In SW-3X1: the CCS zone, about 142 miles of cable

BATTERIES BESIDE GAS

- Batteries take the fast swings; gas runs for days
- PCS skids, MV collection, DC cable in trenches
- In SW-3X1: the battery yard, about 50 miles of cable

Every add-on is more cable: hydrogen, carbon capture and storage each bring their own motors, drives, sensors and controls.

stephan.hardt@southwire.com

UNITS THAT STICK: KW, MW, GW

1 kW

a microwave oven running

kilowatt = 1,000 watts

1 MW

a large store or a small factory

megawatt = 1,000 kW

1 GW

the average use of roughly 800,000 US homes

gigawatt = 1,000 MW

300–550 MW

one large modern gas turbine

the machines in our turbine hall

MW IS POWER

how hard something runs right now: the speedometer

MWH IS ENERGY

power × time: the odometer. 1 MW for 1 hour = 1 MWh

CAPACITY FACTOR

how much a plant actually makes vs. running flat out all year

A 1 GW data-centre campus uses as much power as a city. That is why data centres are pulling new gas plants forward.

Homes: EIA average US residential use, about 10,800 kWh a year (about 1.2 kW on average). Turbine size: typical large-frame H/J-class range.

stephan.hardt@southwire.com

THE WORDS YOU WILL HEAR TODAY

BESS	battery energy storage system
BOP	balance of plant: everything that is not the main machines
BTM	behind the meter: power at the customer's site
COD	commercial operation date: when the plant goes live
EPC	engineering, procurement and construction contractor: builds the plant, buys the cable
FID	final investment decision: the money is committed
GSU	generator step-up transformer
HRSG	heat recovery steam generator: the boiler behind each gas turbine
IPP	independent power producer
ISO / RTO	the grid operator that runs a power market (ERCOT, PJM, MISO)
MCC	motor control centre
POI	point of interconnection: where the plant meets the grid
PPA	power purchase agreement
RICE	reciprocating internal combustion engine: a big gas engine
SCR	selective catalytic reduction: cuts NOx in the HRSG
VFD	variable-frequency drive: controls motor speed

A handout version can go out with the recording.

A MODERN GAS PLANT IS MORE THAN THE POWER BLOCK

PRIVATE 5G / LTE

A monopole with massive-MIMO antennas: the plant's own wireless network.

CABLE

fibre backhaul, tower power, grounding

EV CHARGING + SOLAR CARPORT

Chargers at every bay, a DC fast charger, PV on the carport roof.

CABLE

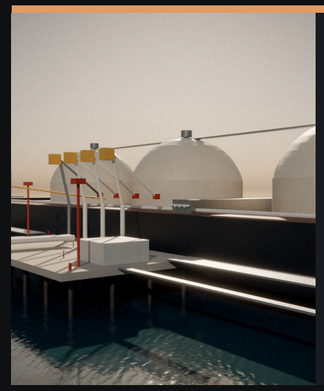
LV feeders, PV wire, a pad-mount transformer

BEHIND-THE-METER DATA CENTRE

A ~150 MW campus fed straight from the plant and the modular yard.

CABLE

34.5 kV collection, cable bus, UPS feeders

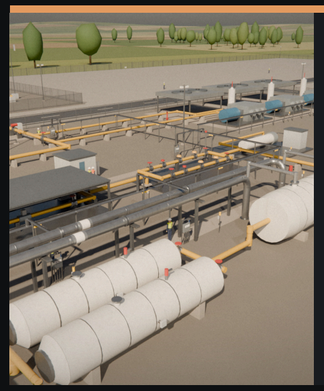


LNG BY SEA

A marine terminal: the carrier at the berth, unloading arms, the tank onshore.

CABLE

hazardous-area cable, long runs to the jetty, MV pumps

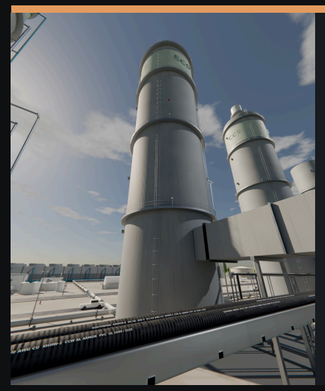


HYDROGEN

Tube-trailer bay and receiving station; an electrolyzer building beside it.

CABLE

Class I Div 2 runs, gas detection, DC cable



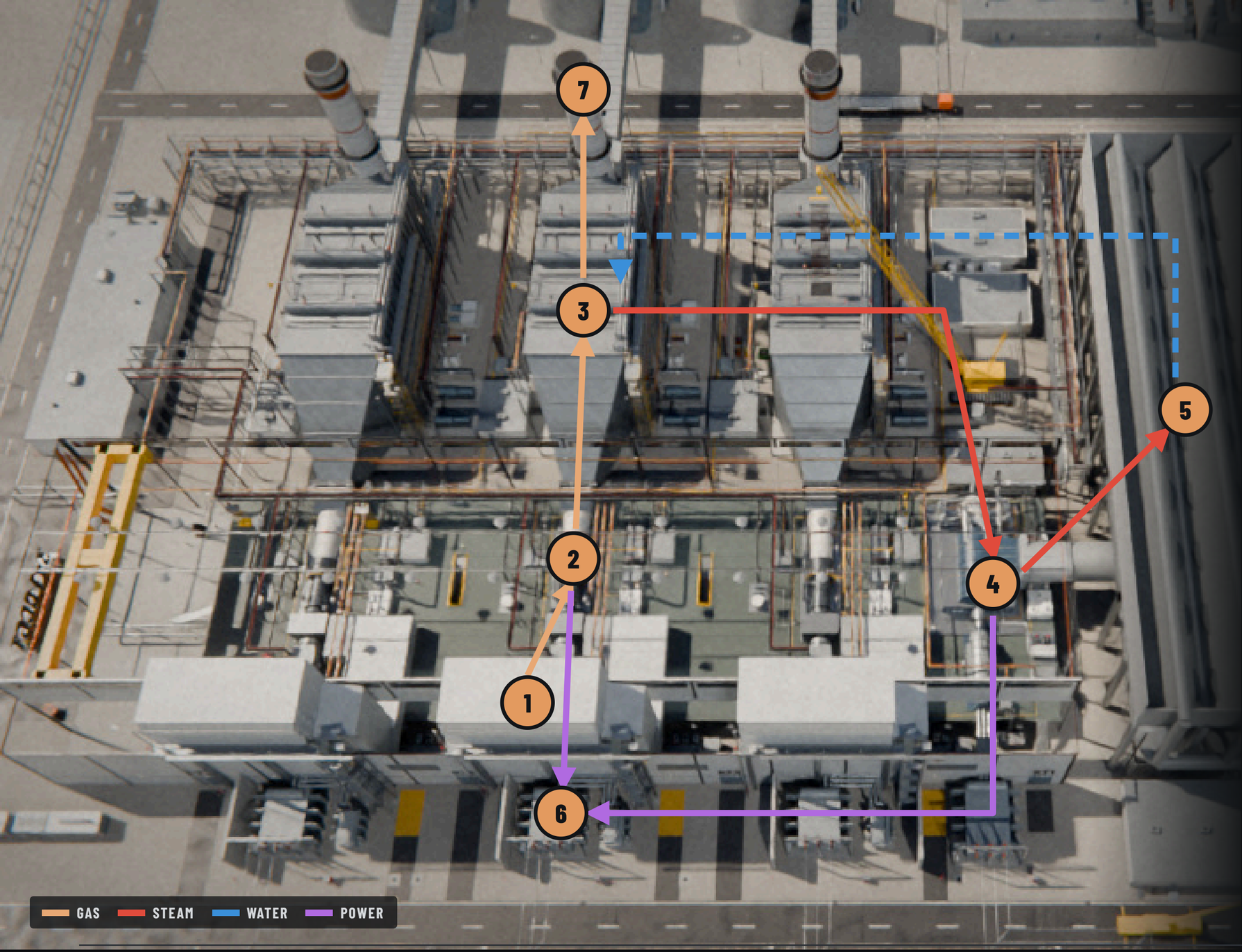
CARBON CAPTURE

Absorber, stripper and compressors on the stack exhaust.

CABLE

a second plant's motors, drives and instruments

All six are in the SW-3X1 model. Each one adds its own network of cable on top of the power block: the campus is the customer, not just the turbines.



FUNDAMENTALS · COMBINED CYCLE

BURN THE GAS ONCE, USE THE HEAT TWICE

1. **AIR AND GAS IN**
   Filters pull in air; gas comes from the gas yard.
   instrumentation, fan and heater power

2. **GAS TURBINE**
   Burns the gas in compressed air; the hot gas spins the turbine and its generator.
   MV-105, iso-phase bus, TC-ER to the skids

3. **HRSG**
   The ~1,100 °F exhaust boils water into steam: no extra fuel.
   thermocouples, MOVs, feed-pump MV

4. **STEAM TURBINE**
   Steam from all three HRSGs spins a fourth generator.
   MV-105, lube-oil and control cable

5. **CONDENSER**
   80 fans cool the spent steam back into water; pumps send it back to the HRSGs.
   VFD cable to every fan

6. **STEP-UP TO THE GRID**
   Each generator's 21 kV goes up to 230 kV in its GSU, then out through the switchyard.
   HV XLPE, control and relay cable

7. **STACK**
   What is left of the exhaust leaves, cleaned by the SCR.
   CEMS instrument cable

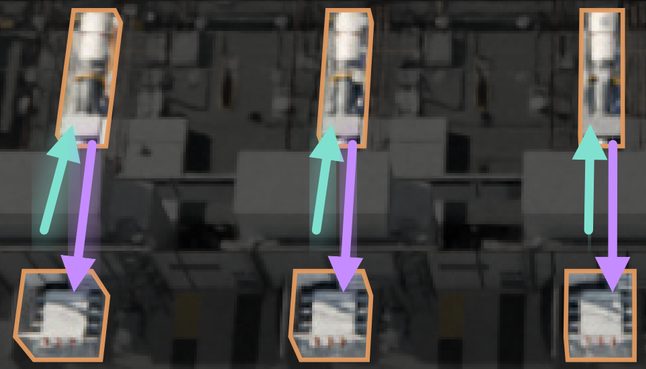
~2/3 OF THE POWER FROM THE GAS TURBINES, ~1/3 FROM THE STEAM: SAME FUEL

stephan.hardt@southwire.com · INSIDE POWER GENERATION 11/16

AIR POWER

THE GAS TURBINE: A JET ENGINE ON THE GROUND

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



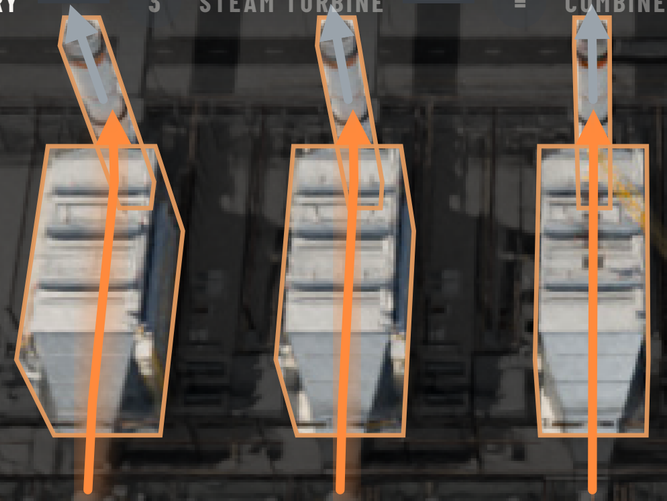
It burns gas in compressed air; the hot gas spins the turbine and its generator. That makes **about two-thirds** of the plant's power, and leaves exhaust at about 1,100 °F.

stephan.hardt@southwire.com

THE HRSG: CATCH THE HEAT

HOT EXHAUST — TO THE STACK

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE

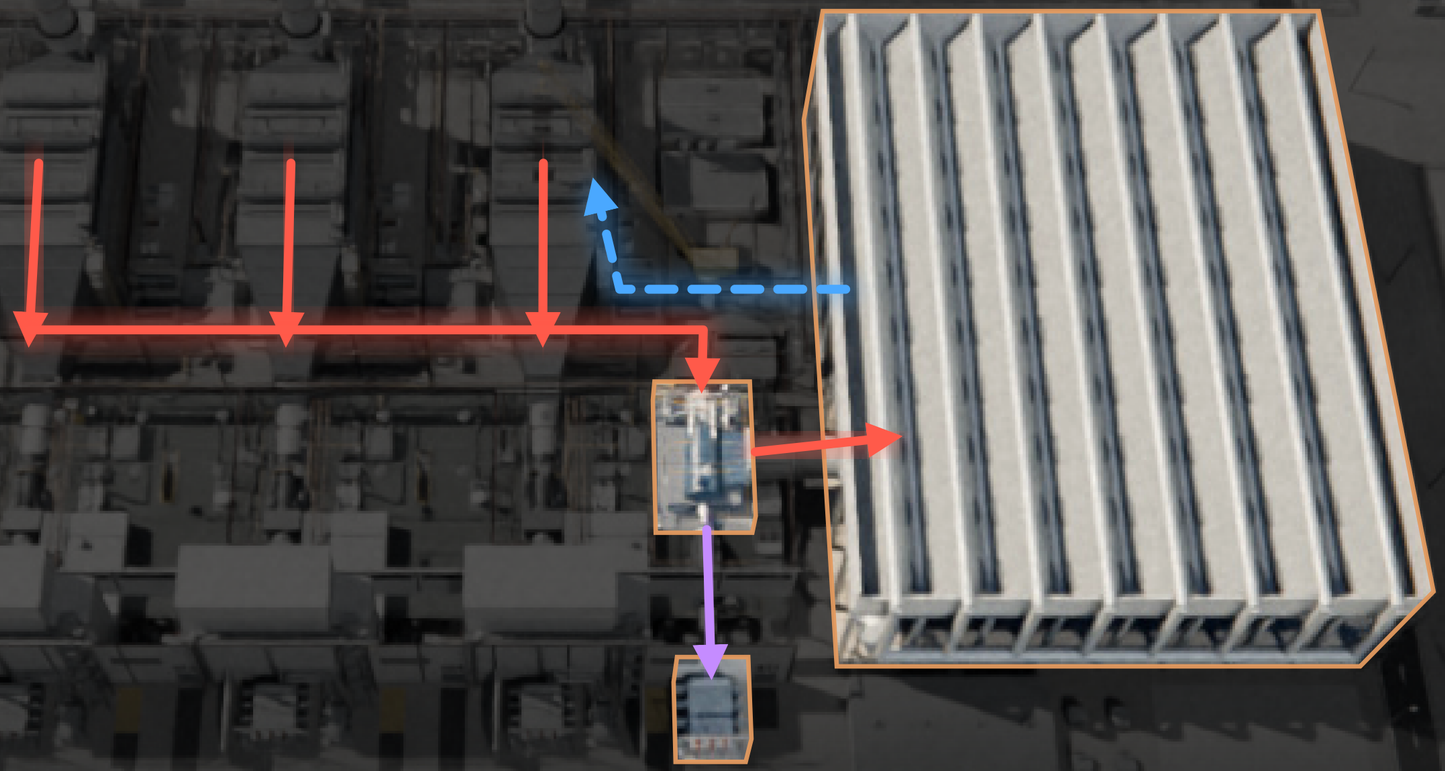


Instead of going up the stack, the hot exhaust runs through a heat-recovery steam generator and boils water into steam. **No extra fuel.**

STEAM WATER POWER

THE STEAM TURBINE: POWER FROM THE LEFTOVERS

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



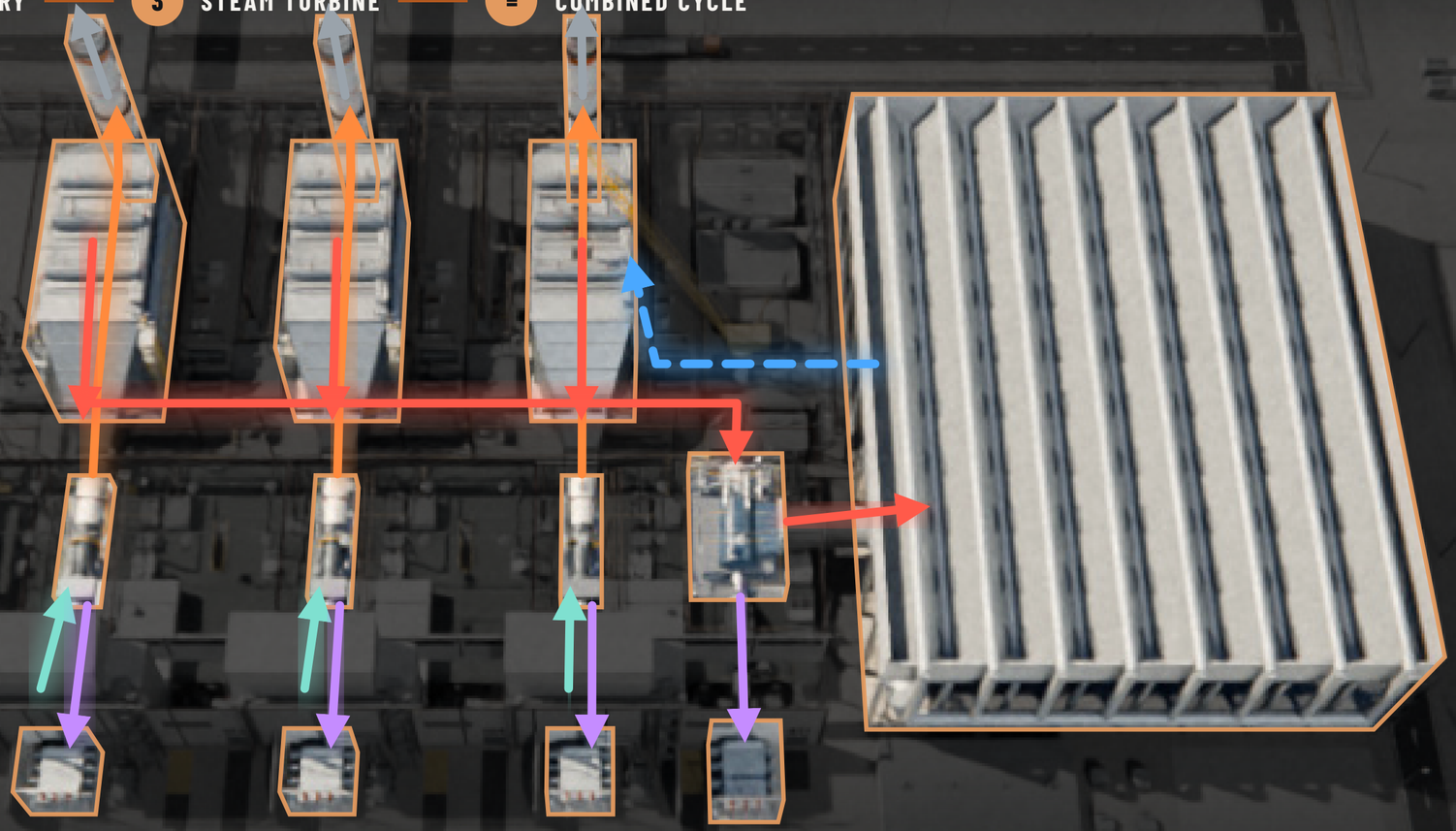
Steam from all three HRSGs spins one more turbine and generator: **about one-third more power.** The air-cooled condenser turns it back into water, and round it goes.

COMBINED CYCLE

BURN THE GAS ONCE, USE THE HEAT TWICE

EXHAUST · STEAM · WATER · POWER

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE — = COMBINED CYCLE

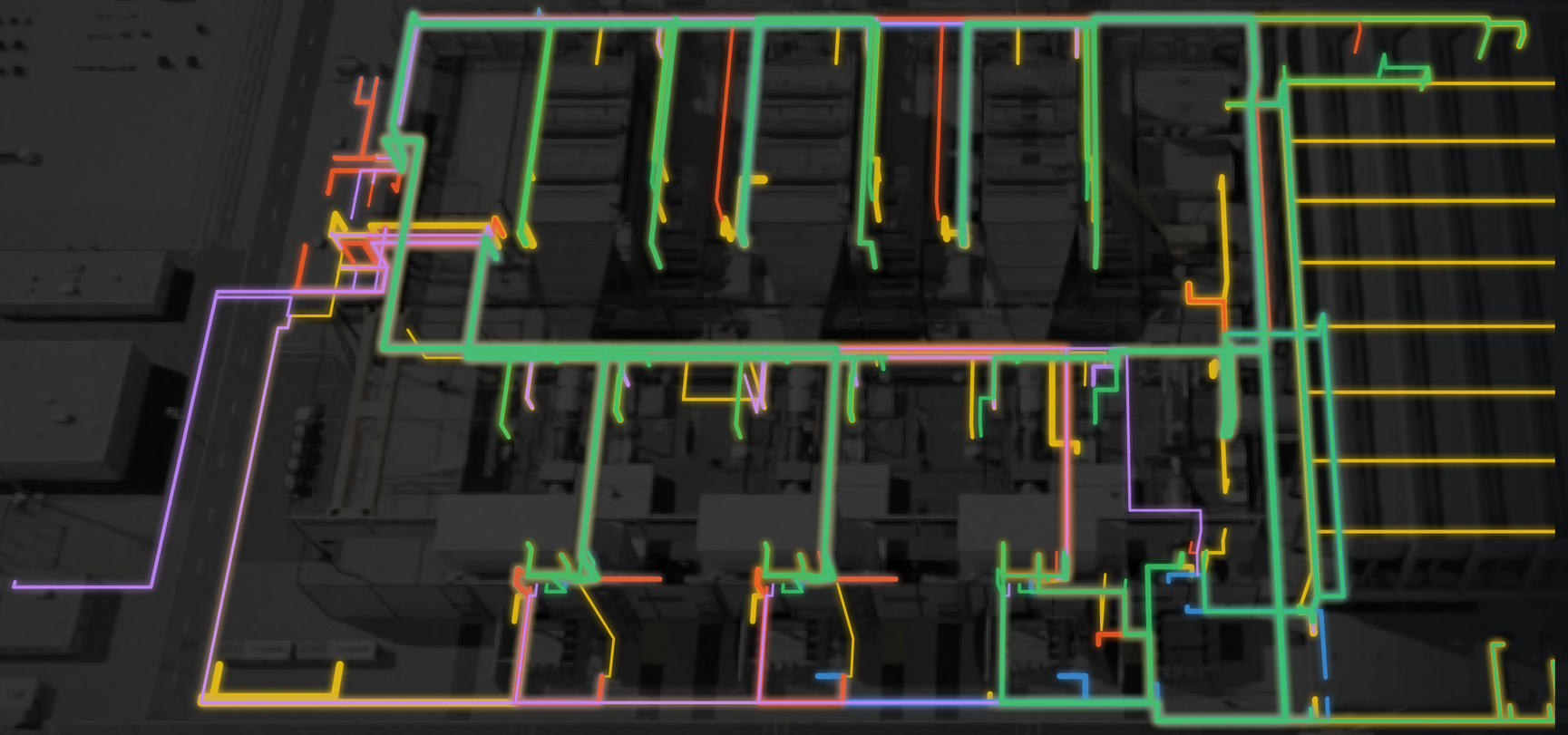


~60%
of the fuel becomes power, vs ~40% for a gas turbine alone

3 + 1
three gas turbines, three HRSGs, one steam turbine: a "3x1"

8,234
cables in this power block and its condenser, in our model

TAKE THE PLANT AWAY: THE CABLE STAYS



THE CABLE IN THIS PICTURE

8,234 cables · ~439 miles

MV / HV POWER — 232 · 28 mi
The big feeders: generators, transformers, the largest motors. MV-105, HV XLPE.

LV POWER — 1,160 · 95 mi
600 V to motors, valves, drives, lighting. TC-ER, ARMOR-X MC-HL, VFD cable.

CONTROL — 1,255 · 94 mi
Starts, stops and trips: breakers, interlocks, MCC starters. Multi-conductor TC-ER.

INSTRUMENTATION — 5,484 · 202 mi
The plant's nerves: pressure, flow, temperature. Shielded pairs, thermocouple extension.

FIBRE AND FIRE ALARM — 103 · 21 mi
The control network, protection and life safety.

The power block and its air-cooled condenser only, from the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.

Every line is a route from the model's cable schedule, in trays, conduit and duct banks; thicker lines carry more cables. **The power block is a web of copper.**

stephan.hardt@southwire.com · INSIDE POWER GENERATION 16/16